

Single-cell-informed transcriptomic transportability of decidual inflammatory macrophage/APC programmes in recurrent pregnancy loss

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Abstract

Recurrent pregnancy loss is a clinically heterogeneous reproductive disorder and a major reproductive-health problem. A substantial fraction of affected patients remain without a definitive explanation after evaluation for chromosomal, anatomical, endocrine, thrombotic and immune contributors, while public transcriptomic studies remain difficult to compare because decidual bulk profiles mix cell abundance, cell state and sampling differences. Here, we used a single-cell-informed transcriptomic transportability design to test whether recurrent-miscarriage-associated tissue-state programmes derived from a disease single-cell/pseudobulk reference recur across independent bulk decidua cohorts. GSE214607 was used to derive compact projection-ready programmes from first-trimester decidua and villi samples. Four signatures were projected into three independent bulk decidua datasets, GSE161969, GSE113790 and GSE178535, using within-cohort standardized expression and mean programme scoring. The primary recurrent-miscarriage decidual inflammatory macrophage/APC programme showed complete 30-gene coverage in all three external cohorts and positive case-associated shifts in each dataset. Cohort-level deltas were +1.019, +0.282 and +0.941, giving a mean delta of +0.747 and positive directional recurrence in 3/3 cohorts. Secondary decidua immune-stromal, villi trophoblast/epithelial and recurrent-miscarriage villi structural/trophoblast programmes showed weaker, modest or context-dependent behaviour. These findings support exploratory programme-level transportability of an inflammatory decidual macrophage/APC tissue-state axis in recurrent pregnancy loss. The result is observational and hypothesis-generating; it does not establish causality, diagnostic performance, clinical utility or standalone biomarker validation.

Introduction

Infertility and pregnancy loss impose substantial clinical, psychological and societal burdens. The World Health Organization estimates infertility as a common global reproductive-health condition, while recurrent pregnancy loss (RPL) represents a distinct clinical problem requiring its own evaluation and management pathway [1-3]. Contemporary guidelines differ in definitions and thresholds, but both ESHRE and ASRM emphasize the heterogeneity of RPL and the need to evaluate established contributors such as genetic abnormalities, uterine factors, endocrine disease, antiphospholipid syndrome and other maternal or embryonic causes [2,3]. Beyond these recognized categories, miscarriage and recurrent miscarriage also carry physical, psychological and future obstetric consequences, making mechanistic understanding of early pregnancy failure a human-health priority rather than a narrow fertility issue [4,5].

The early maternal-fetal interface provides a biologically plausible site for unexplained or incompletely explained pregnancy loss. Human decidualization transforms endometrial stromal fibroblasts into specialized decidual cells, coordinates glandular and vascular remodelling, and creates a tissue matrix in which immune competence must coexist with tolerance of fetal trophoblast [6,7]. Single-cell atlases of first-trimester decidua and placenta have shown that this interface is a structured multicellular ecosystem containing decidual stromal cells, macrophages, natural killer cells, T cells, endothelial and mural cells, epithelial compartments and invading trophoblasts [8,9]. Trophoblast-decidual immune crosstalk, local trophoblast recognition by uterine immune cells and regulatory T-cell programmes are therefore central to early placentation and immune balance [10-13].

RPL has repeatedly been linked to immune dysregulation, but the direction, cellular basis and translational meaning of immune signals remain unresolved. Single-cell and immune-profiling studies have reported altered distributions of decidual leukocyte subsets in RPL, unexplained RPL and recurrent miscarriage, including changes in NK-cell, T-cell and myeloid compartments [14-17]. Decidual macrophages are particularly relevant because they participate in phagocytosis, tissue remodelling, angiogenesis, trophoblast interaction, cytokine production and immune regulation [18,22,23]. Disease-focused studies have associated RPL or recurrent spontaneous abortion with altered decidual macrophage states, CCR1+ macrophage dysfunction, M1-like polarization and macrophage-related diagnostic signatures [18-21,30]. Complement-associated macrophage biology is also plausible at the maternal-fetal interface because C1q and complement-related pathways contribute to implantation-site inflammation, trophoblast invasion and tissue remodelling, although excessive or mislocalized complement activity can be harmful [24-27].

Despite this biological rationale, public transcriptomic evidence remains hard to integrate. Bulk decidual RNA profiles can detect disease-associated expression differences, but they cannot directly distinguish increased macrophage abundance from macrophage activation, altered decidual stromal composition, trophoblast admixture, tissue injury or platform-specific artefact. Many public RPL studies are small, and single-cohort differential-expression lists may not transport across datasets [31-36]. Single-cell-informed projection offers an intermediate strategy: derive interpretable programmes from a cellular reference, then test whether those programmes recur directionally across independent bulk cohorts. This approach is different from diagnostic classifier construction; it evaluates whether a biologically coherent tissue-state axis is detectable beyond the discovery dataset.

The present study addresses this gap by testing the transportability of GSE214607-derived recurrent-miscarriage and tissue-context programmes across three independent bulk decidua cohorts. The aims were: first, to derive compact and interpretable projection-ready programmes from a disease single-cell/pseudobulk reference; second, to determine whether these programmes show adequate gene coverage when projected into independent bulk decidua datasets; and third, to assess whether a recurrent-miscarriage decidual inflammatory macrophage/APC programme shows concordant case-associated recurrence across cohorts. Figure 1 summarizes the workflow, Table 1 summarizes the datasets, and Supplementary Table S1 documents metadata inclusion.

Materials and Methods

This was a retrospective computational reanalysis of public de-identified transcriptomic datasets. No new participants were recruited, and no new identifiable human data were generated. The analysis was designed around programme transportability rather than biomarker discovery or diagnostic classification. The overall workflow is summarized in Fig. 1.

GSE214607 was used as the primary disease single-cell reference because it contains first-trimester decidua and villi samples from control and recurrent-miscarriage groups and was originally generated to explore

macrophage-associated recurrent miscarriage signals [28-30]. Raw 10x Genomics matrix files were imported as sparse count matrices, checked for matrix/barcode/feature consistency and assigned to control decidua, recurrent-miscarriage decidua, control villi or recurrent-miscarriage villi categories. Per-cell quality-control metrics included total counts and detected genes. Cells passing predefined feature thresholds were retained for sample-level aggregation.

To avoid pseudoreplication, retained GSE214607 cells were aggregated into sample-level pseudobulk profiles by summing counts within each sample. Library-size-normalized log2 counts per million were then calculated. Pseudobulk aggregation was used for programme derivation because cells from the same individual are not independent biological replicates and single-cell differential-expression procedures can underestimate variance if sample structure is ignored [37,38]. The pseudobulk profiles were used only to derive exploratory programmes and not to make formal differential-expression claims.

Candidate programmes were derived from three contrast families: recurrent-miscarriage decidua versus control decidua, recurrent-miscarriage villi versus control villi and decidua versus villi. Candidate genes were ranked by direction and magnitude of expression shift. Lists were then filtered for interpretability: low-information noncoding, housekeeping, ribosomal, mitochondrial and technical features were excluded unless they had explicit maternal-fetal-interface relevance. The final projection-ready framework retained four 30-gene signatures: a recurrent-miscarriage decidual inflammatory macrophage/APC programme, a decidua immune-stromal identity programme, a villi trophoblast/epithelial identity programme and a recurrent-miscarriage villi structural/trophoblast programme (Table 2; Supplementary Table S2).

Three external bulk decidua datasets were used for projection: GSE161969, GSE113790 and GSE178535 [31-36]. GSE161969 was processed from a public gene-symbol FPKM matrix and transformed as $\log_2(\text{FPKM} + 1)$. GSE113790 was imported as a processed FPKM expression matrix and curated at the gene-symbol level. GSE178535 was analysed from its public normalized log2 expression matrix. Sample labels were manually curated to harmonize control and RPL/RSA case categories while preserving dataset-specific terminology (Supplementary Table S1).

For each cohort and signature, genes were intersected with the cohort expression matrix, and coverage was recorded before scoring. Within each cohort, each gene was standardized across samples. Programme scores were then calculated as the mean standardized expression of the matched signature genes. This simple mean-z scoring was chosen to preserve interpretability and avoid cross-platform absolute-expression assumptions; related single-sample gene-set approaches use analogous logic to summarize coordinated expression at the sample level [39].

Transportability was evaluated using technical coverage, case-minus-control programme-score deltas and directional concordance. Complete or near-complete signature coverage was required for confident interpretation. For each signature and dataset, mean scores were calculated for controls and cases, and case-minus-control deltas were computed. Directional recurrence was defined as a consistent sign of case-associated shift across cohorts. Wilcoxon rank-sum tests and Benjamini-Hochberg-adjusted values were calculated descriptively but were not used as the primary criterion because the external cohorts were small [40]. Full projection-score and metadata files are listed in Supplementary Tables S2-S4.

Results

The analysis integrated one disease single-cell/pseudobulk reference and three independent bulk decidua cohorts. GSE214607 provided the programme-derivation reference, while GSE161969, GSE113790 and GSE178535 provided independent projection settings (Table 1; Fig. 1). The use of three external cohorts

allowed the study to prioritize directional recurrence across independent datasets rather than within-dataset statistical significance.

Pseudobulk contrasts from GSE214607 generated both recurrent-miscarriage-associated and tissue-context programmes. After interpretability filtering, four compact signatures were retained (Table 2). The recurrent-miscarriage decidual inflammatory macrophage/APC programme was designated the primary programme because it was derived from the disease-associated decidual contrast and included coherent myeloid, antigen-presentation and complement-associated genes. The signature included HLA-DRA, HLA-DPA1, HLA-DPB1, CD74, S100A8, S100A9, AIF1, MS4A7, C1QA, C1QB, C1QC, CTSZ, SPP1 and GRN, supporting interpretation as a macrophage/APC-rich inflammatory tissue-state programme rather than an arbitrary gene list (Table 2; Supplementary Table S2; Fig. 2).

The primary macrophage/APC programme showed complete technical coverage in all external bulk cohorts. All 30 genes were detected in GSE161969, GSE113790 and GSE178535, supporting reliable projection across different processed expression matrices. Secondary signatures also had high coverage, but the macrophage/APC programme was the most technically robust primary disease-associated signature (Supplementary Table S3; Supplementary Fig. 1).

The macrophage/APC programme showed a positive case-associated shift in all three external cohorts. The case-minus-control delta was +1.019 in GSE161969, +0.282 in GSE113790 and +0.941 in GSE178535. This yielded positive directional recurrence in 3/3 cohorts, a mean delta of +0.747 and a median delta of +0.941 (Table 3; Fig. 3). Because the cohorts were small, this result is interpreted as exploratory directional recurrence rather than a validated diagnostic effect. Nevertheless, the consistency of direction across three datasets supports programme-level transportability of the decidual inflammatory macrophage/APC axis.

Secondary signatures showed weaker or context-dependent behaviour. The decidua immune-stromal identity programme was positive in all three cohorts but with smaller deltas (+0.573, +0.206 and +0.031). The villi trophoblast/epithelial identity programme showed modest positive shifts (+0.134, +0.406 and +0.156), while the recurrent-miscarriage villi structural/trophoblast programme was mixed (-0.097, +0.245 and -0.007). Thus, not all GSE214607-derived signatures showed the same behaviour in decidua-focused external cohorts (Table 3; Fig. 4). This pattern supports tissue-context-aware interpretation and argues against a nonspecific projection artefact shared equally by all signatures.

Together, the results identify a single-cell-derived recurrent-miscarriage decidual inflammatory macrophage/APC programme with complete 30-gene coverage and positive directional recurrence across three independent bulk decidua cohorts. The main result is a reproducible tissue-state signal, not a causal proof, clinical diagnostic model or biomarker validation. Intermediate single-cohort and two-cohort analyses are retained for reproducibility in Supplementary Fig. 2.

Discussion

This study provides exploratory evidence that a GSE214607-derived recurrent-miscarriage decidual inflammatory macrophage/APC programme is directionally transportable across three independent bulk decidua cohorts. The study was designed to answer a conservative reproducibility question: does an interpretable single-cell-derived tissue-state programme recur across datasets, tissues and preprocessing contexts? The answer was strongest for the macrophage/APC programme, which showed complete gene coverage and positive case-associated shifts in GSE161969, GSE113790 and GSE178535. This directly addresses the first and second aims by demonstrating that compact programme derivation and technically robust projection are feasible, and addresses the third aim by showing concordant cross-cohort recurrence of the primary inflammatory myeloid programme.

Scientifically, the result fits a growing literature in which RPL is viewed not only as an embryonic or anatomical problem but also as a disorder that may involve disturbed decidual immune balance. Guidelines emphasize that RPL is clinically heterogeneous and that many cases remain unexplained after standard evaluation [2,3]. The Lancet miscarriage series further frames recurrent miscarriage as a high-burden condition that requires better biological understanding and improved care pathways [4,5]. The present data do not redefine RPL clinically, but they add a reproducible transcriptomic tissue-state observation that may help organize future mechanistic work.

The macrophage/APC programme is biologically plausible because macrophages are positioned at the interface between immune tolerance, tissue repair, trophoblast interaction, angiogenesis and inflammation. Single-cell studies of normal early pregnancy show that decidua contains diverse immune and stromal populations whose interactions are essential for successful placentation [8-11]. Disease-focused single-cell and immune-profiling studies have independently reported altered decidual leukocyte states in RPL or unexplained RPL, including myeloid, NK-cell and T-cell alterations [14-17]. The current finding is therefore similar in direction to published immune-landscape studies: it supports the presence of an altered decidual immune tissue state. Its specific contribution is methodological and cross-dataset: it shows that a compact macrophage/APC programme can be projected and recovered across three independent bulk cohorts rather than inferred from one dataset alone.

The programme also agrees with macrophage-focused RPL literature, but it should be interpreted carefully. Reviews and primary studies have emphasized that decidual macrophages cannot be reduced to a simple M1/M2 dichotomy and may adopt hybrid, context-dependent states [18,22,23]. CCR1+ macrophage dysfunction, M1-like macrophage polarization and impaired anti-inflammatory macrophage programming have been reported in RPL or recurrent spontaneous abortion contexts [19-21]. The present signature includes antigen-presentation genes, inflammatory myeloid markers and C1q-associated macrophage genes, so it is coherent with these reports. However, the bulk projection design cannot determine whether the signal reflects more macrophages, a different macrophage state, increased antigen presentation, complement-associated tissue remodelling or secondary response to tissue injury. This is a key difference from functional macrophage studies: our analysis supports recurrence of a programme, not the mechanism that generates it.

Complement-related genes in the primary signature, especially C1QA, C1QB and C1QC, add biological interest. Complement pathways are active at the implantation site and can contribute to inflammation, trophoblast invasion, vascular remodelling and clearance of apoptotic material [24-26]. Recent decidual bioinformatics work has also linked complement/coagulation modules and macrophage abundance to RPL datasets [27]. The similarity is useful because it places the current C1q-containing macrophage/APC programme within a broader complement-inflammation literature. The difference is equally important: the current study does not claim that complement activation causes RPL or that C1q is independently pathogenic. C1q genes may mark macrophage identity, trophoblast/stromal interaction, tissue-remodelling macrophages or local innate immune activation, and these possibilities require spatial and protein-level validation.

Methodologically, the study contributes a transportability-oriented way to reuse public reproductive transcriptomic data. Single-cell datasets provide cellular resolution but are often small, whereas bulk datasets provide independent cohorts but have mixed-cell composition. Pseudobulk summarization respects biological replication in single-cell analysis and reduces false confidence caused by treating cells as independent samples [37,38]. Mean standardized programme scoring is transparent and interpretable, and the use of directional concordance across three cohorts is appropriate for small public datasets where individual P values are unstable [39,40]. This differs from diagnostic classifier studies, including the original macrophage-marker model associated with GSE214607 [30]. The present analysis deliberately avoids claiming a predictive model and instead asks whether the biological programme itself is transportable.

The weaker behaviour of the secondary signatures is informative. If every GSE214607-derived signature shifted similarly across all external datasets, the result could reflect a nonspecific batch or disease-projection artefact. Instead, the decidua immune-stromal programme was positive but weaker, villi/trophoblast-related programmes were modest or mixed, and the macrophage/APC signature was the most consistent. This supports the interpretation that decidua-relevant inflammatory myeloid biology, rather than generic signature projection, drives the primary result. It also reinforces the importance of tissue context: a villi-derived programme should not necessarily transport strongly into decidua-enriched bulk cohorts.

Translationally, the finding should be treated as a prioritization result. It nominates a decidual inflammatory macrophage/APC axis for future validation in larger, better annotated cohorts. Useful follow-up studies would include euploid versus aneuploid loss stratification, gestational-age matching, standardized tissue sampling, histopathology, fetal karyotype, endocrine and immune metadata, and prospective replication. Spatial transcriptomics, multiplex immunofluorescence, imaging mass cytometry or targeted immunohistochemistry could test whether the programme localizes to specific decidual niches and whether it reflects macrophage abundance, activation state, antigen-presentation activity, complement-associated remodelling or interaction with trophoblast and stromal compartments.

Several limitations remain. First, the external datasets are small, so the evidence rests on directionality across cohorts rather than cohort-level statistical validation. Second, public metadata are incomplete, and clinical labels such as RPL, RM, RSA and unexplained RSA may not represent identical biological entities. Third, bulk programme scores cannot distinguish cell abundance from cell state. Fourth, GSE214607 itself is small and disease-enriched, so derived signatures may capture disease-context programmes rather than universal macrophage identities. Fifth, the reference list and biological framing in v2 are broader than v1.1, but the central empirical result is unchanged: a 30-gene macrophage/APC programme shows positive recurrence across three cohorts.

In conclusion, this study identifies a single-cell-derived recurrent-miscarriage decidual inflammatory macrophage/APC programme that recurs directionally across three independent bulk decidua datasets. The revised evidence framework positions the result within RPL clinical burden, maternal-fetal-interface biology, decidual macrophage literature, complement-associated inflammation and public-data transportability methodology. The finding is biologically coherent and methodologically useful, but it remains exploratory and should not be interpreted as causal, diagnostic or clinically actionable without independent spatial, protein-level and prospective validation.

Conclusion

This study provides exploratory evidence that a single-cell-derived recurrent-miscarriage decidual inflammatory macrophage/APC programme recurs directionally across three independent bulk decidua transcriptomic cohorts. The finding supports the use of single-cell-informed transportability analysis to identify reproducible tissue-state programmes in recurrent pregnancy loss. The macrophage/APC axis is biologically coherent and methodologically transportable, but it should not yet be considered causal, diagnostic or clinically actionable.

Data Availability Statement

Figure-caption support files, reproducibility-audit outputs and release documentation supporting manuscript assembly are summarized in Supplementary Table S4.

All datasets analysed in this study are publicly available from GEO or associated public sequence repositories under accession numbers GSE214607, SRP400574, GSE161969, SRP293708, GSE113790, SRP143517 and

GSE178535. Derived signatures, curated metadata summaries, projection scores, transportability summaries, figures, manuscript drafts, QC reports and reproducibility documentation are available at <https://github.com/rmaghembe1/rpl-decidua-transportability>. Large raw downloads and regenerable intermediate objects are not redistributed because the source datasets remain publicly available.

Author contributions

Reuben S. Maghembe: conceptualisation, methodology, software, formal analysis, investigation, data curation, visualisation, writing - original draft, writing - review and editing, project administration.

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Competing interests

The author declares no competing interests.

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Supporting information captions

Supplementary Table S1. Dataset metadata and sample inclusion framework. Curated sample metadata, tissue source, disease group, analysis role and inclusion/exclusion status for GSE214607, GSE161969, GSE113790 and GSE178535.

Supplementary Table S2. GSE214607-derived projection-ready signature gene sets. Full gene lists for the four projection-ready signatures.

Supplementary Table S3. Three-cohort signature projection and transportability statistics. Signature scores, gene-overlap metrics, cohort-specific case-minus-control deltas and cross-cohort concordance summaries.

Supplementary Table S4. Figure-caption support and reproducibility audit. Figure-source tables, script/output inventory, release-manifest notes and interpretation-boundary documentation.

Supplementary Figure S1. Dataset processing, quality-control and signature-coverage summary.

Supplementary Figure S2. Intermediate single-cohort and two-cohort projection summaries supporting the final three-cohort analysis.

Figure legends

Figure 1. Study design and analytical workflow

Single-cell-informed transcriptomic transportability workflow for recurrent pregnancy loss decidua analysis. GSE214607 was used as a disease single-cell/pseudobulk reference and GSE161969, GSE113790 and GSE178535 were used as independent bulk decidua projection cohorts. Candidate programmes were derived from pseudobulk contrasts, filtered for interpretability and projected into each bulk cohort using within-cohort standardized expression. Transportability was evaluated by coverage, case-minus-control deltas and directional concordance.

Figure 2. GSE214607-derived projection-ready tissue-state signatures

Derivation of projection-ready recurrent-miscarriage and tissue-context programmes from GSE214607. Four compact signatures were retained: recurrent-miscarriage decidual inflammatory macrophage/APC up, decidua immune-stromal identity, villi trophoblast/epithelial identity and recurrent-miscarriage villi structural/trophoblast.

Figure 3. Three-cohort transportability of the macrophage/APC programme

The recurrent-miscarriage decidual inflammatory macrophage/APC programme showed complete 30/30 gene coverage and positive case-associated shifts in GSE161969, GSE113790 and GSE178535. Deltas were +1.019, +0.282 and +0.941, respectively.

Figure 4. Cross-cohort delta summary for all projected signatures

Case-minus-control programme-score deltas for four GSE214607-derived signatures across external cohorts. The macrophage/APC programme showed the strongest and most consistent positive directional recurrence; villi/trophoblast-associated programmes were weaker or mixed.

Tables

Table 1. Public datasets used in the analysis

Dataset	Assay/data type	Tissue/source	Groups used	Role in study	Key processing	Main limitation
GSE214607	Single-cell RNA-seq / pseudobulk reference	First-trimester decidua and villi	Control and recurrent miscarriage	Disease single-cell reference; programme derivation	Raw 10x matrices imported, QC-filtered and aggregated to pseudobulk profiles	Limited sample numbers; exploratory programme derivation
GSE161969	Bulk RNA-seq processed expression	Decidua	Control and RPL	External projection cohort	Gene-symbol FPKM matrix; log2(FPKM + 1); within-cohort scoring	Very small cohort; limited metadata
GSE113790	Bulk transcriptomic processed expression	Decidua	Control and RSA	External projection cohort	FPKM matrix imported; labels curated; within-cohort scoring	Very small cohort; metadata harmonization constraints
GSE178535	Bulk transcriptomic processed expression	Decidua	Control and RSA	External projection cohort	Public normalized log2 matrix; labels curated; within-cohort scoring	Very small cohort; platform/preprocessing differences

Table 2. GSE214607-derived projection-ready signatures

Programme ID	Short label	Source contrast	Genes	Biological interpretation	Use
GSE214607_RM_decidua_inflammatory_macrophage_APC_up	RM decidua inflammatory macrophage/APC up	RM decidua vs control decidua	30	Antigen-presentation, inflammatory myeloid and complement-associated macrophage/APC programme	Primary programme
GSE214607_decidua_immune_stromal_identity_up	Decidua immune-stromal identity	Decidua vs villi	30	Decidua-enriched immune and stromal tissue-context programme	Secondary comparator
GSE214607_villi_trophoblast_epithelial_identity_u	Villi trophoblast/epithelial identity	Villi vs decidua	30	Villi/trophoblast/epithelial tissue-	Tissue-context comparator

Programme ID	Short label	Source contrast	Genes	Biological interpretation	Use
p				context programme	
GSE214607_RM_villi_structural_trophoblast_down	RM villi structural/trophoblast programme	RM villi vs control villi	30	Villi structural/trophoblast-associated programme	Secondary tissue-context comparator

Table 3. Three-cohort transportability summary

Programme	GSE161969 delta	GSE113790 delta	GSE178535 delta	Positive cohorts	Minimum coverage	Concordance interpretation
RM decidua inflammatory macrophage/A PC up	+1.019	+0.282	+0.941	3/3	100%	Concordant positive directional signal
Decidua immune-stromal identity	+0.573	+0.206	+0.031	3/3	96.7%	Positive but weaker/context-dependent
Villi trophoblast/epithelial identity	+0.134	+0.406	+0.156	3/3	100%	Modest positive but tissue-context dependent
RM villi structural/trophoblast programme	-0.097	+0.245	-0.007	1/3	96.7%	Mixed/weak; not transportable in decidua cohorts